Kaustav Saha

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PROFESSIONAL SUMMARY

Computer Science Engineering student with a strong academic record and hands-on programming experience, applying object-oriented programming and data structures & algorithms to build real-world software systems. Experienced in end-to-end software development, from deep learning pipelines to multi-agent architectures with exposure to both front-end and back-end technologies. Skilled in writing clean, modular, version-controlled code and quickly adapting to new tools, frameworks, and problem domains. Eager to apply strong analytical and problem-solving abilities to deliver scalable, high-quality software solutions in a fast-paced, collaborative environment.

EDUCATION

Institute of Engineering and Management — Kolkata, India
B.Tech : Computer Science & Engineering (IoT and Cyber Security including Blockchain Technology) – CGPA : 9.18 — 2023 – 2027

Sudhir Memorial Institute — Kolkata, India
Class 12, CBSE – 73.3% — 2023

Sudhir Memorial Institute — Kolkata, India
Class 10, CBSE – 84% — 2021

EXPERIENCE

Research Intern – Software & AI Engineering — May 2026 – July 2026
Indian Institute of Technology (IIT), Patna

- **Deep Learning Model:** Developed a PyTorch-based Hierarchical Multi-Label Text Classification (HMTC) system to resolve high-dimensional, multi-tiered data taxonomies while tracking multi-variable evaluation metrics.
- **Data Ingestion Pipelines:** Built automated text preprocessing and tokenization workflows to standardize raw textual data formats prior to model execution.
- **Collaboration & Iteration:** Collaborated with fellow interns and researchers within a cooperative team environment to define project scope and evaluation approach, incorporating feedback through iterative rounds of model refinement and benchmarking.
- **SDE Design Principles:** Maintained a clean, modular codebase using Git/GitHub for version control, applying internal best practices to improve the efficiency, reliability, and observability of model evaluation pipelines at scale.

SKILLS

Languages	C++, Python, SQL, C
Web Development	React.js, Node.js, Express, Flask, Vite, Tailwind CSS, REST APIs
Databases	PostgreSQL, Drizzle ORM, Firebase, FAISS
AI/ML	PyTorch, LangChain, LangGraph, RAG, Pandas, NumPy
Tools	VS Code, Git, GitHub, Google Colab, Kaggle, Vitest
Core Concepts	Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems, Machine Learning, Cybersecurity, Blockchain Fundamentals
Management	Coordinated large-scale hackathon events – logistics, venue, and scheduling

RESEARCH WORK & PUBLICATIONS

A Multi-Agent AI Architecture for Personalized Medication Management – Presented at the 4th International Conference on Computational Intelligence, Data Science and Cloud Computing (ICDC 2026); currently in press with Springer Nature.

PROJECTS

Multi-Agent AI Architecture – Personalized Medication Management

- **Tech Stack:** Python, LangChain, LangGraph, Multi-Agent Systems, RAG, Gemini 2.5 Flash, Faker, FAISS, Sentence-Transformers, Streamlit, Pandas, NumPy, Pydantic
- Orchestrated a 3-agent autonomous system – comprising Monitor, Nurse, and Medication Manager, to analyze adherence trends, optimizing the shift from automated reminders to personalized care.
- Total system latency of 2.5 seconds (actual processing) with Medication Manager queries at 2.1s, Nurse Agent reminders at 1.5s, and adherence coaching at 2.3s.

CuraCore HealthBot – RAG-based Web Application — GitHub: github.com/BlackClover-pvt/CuraCore

- **Tech Stack:** Python, Flask, LangChain, Google Gemini API, FAISS, Firebase, Google Maps API, React.js, Tailwind CSS, Vite
- Built a conversational AI to execute precise, real-time Retrieval-Augmented Generation (RAG) over unstructured medical texts with vector similarity search and context memory management.
- Implemented vector similarity search and context memory management, deploying the end-to-end web application.

BugX – AI-Powered Code Review Platform — GitHub: github.com/BlackClover-pvt/BugX

- **Tech Stack:** React 19, Node.js, Express, TypeScript, PostgreSQL, Drizzle ORM, Google Gemini AI
- Built a full-stack automated code review web application with an interactive in-browser code editor and real-time AI-powered defect detection, security audits, and code optimization using Google Gemini (gemini-2.5-flash).
- Implemented a relational persistence layer with PostgreSQL and Drizzle ORM for user accounts, reviews, and detailed findings, with authentication using Argon2id password hashing, Zod schema validation, and persistent HTTP-only cookie sessions via connect-pg-simple.
- Designed a responsive dark glassmorphic UI that gates the workspace behind a dedicated authentication entry flow.

CERTIFICATIONS

- Google Cloud Computing Foundations – Google, NPTEL
- The Joy of Computing using Python – IIT KGP, NPTEL
- Fundamentals of OOP – IIT Roorkee, NPTEL
- Cyber Security Fundamentals – University of London, Coursera
- Foundations of Cryptography – IIIT Bangalore, NPTEL
- Data Analytics with Python – IIT Roorkee, NPTEL

PARTICIPATION & LEADERSHIP

- **Event Coordinator** – IEM Hackoasis 2.0 — 2025
  - Managed end-to-end logistics, venue allocation, and operational scheduling for large-scale hackathon teams.
- **Conference Volunteer** – 4th International Conference on Computational Intelligence, Data Science and Cloud Computing (ICDC 2026) — 2026
- **IEEE CIS Summer School 2026** – Participated in "Computational Intelligence in the Age of LLMs and LVMs: Foundations, Challenges, and Applications," IIT Patna. 2026